The Consumption Stimulus Effects of Student Debt Forgiveness Policies*

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Abstract

The share of liquidity-constrained households holding student debt more than doubled between 2001 and 2016, and student debt relief has become a central policy question. We study student debt forgiveness ~~relief~~ in a general equilibrium model with the main institutional features of federal student loan repayment. The model matches the marginal propensity to consume out of payment reductions estimated for recent forgiveness programs. We find that forgiveness poses a tradeoff between stimulating the consumption of forgiven borrowers and raising welfare. ~~consumption stimulus and welfare.~~ Concentrating forgiveness on low-income borrowers, who are the most likely to be constrained, raises welfare but lowers the consumption response ~~consumption stimulus~~, while cash transfers to the same borrowers raises both ~~do not~~. Income-driven repayment (IDR) plans and delinquency explain why. Forgiveness does little to lower the payments of borrowers on IDR plans or in delinquency, who are concentrated at the bottom of the earnings distribution. ~~Borrowers on IDR plans or in delinquency do not see a lower payment when their balance falls, and they are concentrated at the bottom of the earnings distribution.~~ Removing IDR plans eliminates the tradeoff, while removing delinquency does not. ~~The aggregate response also depends on how forgiveness is financed, as with other fiscal transfers.~~

*The views in this paper do not represent the views of the Bank of Canada.

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1 Introduction

"This is the single most effective executive action available for a massive economic stimulus."

— Elizabeth Warren, U.S. Senator from Massachusetts, November 2020

"Student loan debt forgiveness likely has a multiplier close to zero."

— Jason Furman, former Chair, Council of Economic Advisers, November 2020

The share of liquidity-constrained households that hold student debt more than doubled between 2001 and 2016, from 12% to 25%, nearly matching the share that hold mortgage debt (27% in 2016). These borrowers are at the center of a contentious debate about whether student debt relief would stimulate consumption. ~~the economic effects of student debt relief.~~ Proponents argue that debt forgiveness would not only help borrowers but also stimulate the economy, while critics contend that the stimulus would be small. Standard reasoning suggests the answer turns on whether forgiven borrowers are liquidity constrained, because constrained households consume a large fraction of any increase in disposable income.

The debate raises two questions. How much does student debt forgiveness raise the consumption of forgiven borrowers, and how does the design of forgiveness shape this consumption response and welfare? ~~What are the aggregate consumption effects of student debt relief, and how does the design of relief shape those effects?~~ Dinerstein et al. (2025) provide evidence on the first question. They compare borrowers whose debt was canceled between 2021 and 2024 with borrowers who had not yet received forgiveness, and estimate the effect of forgiveness on their consumption. Their estimates, however, apply to the specific groups of borrowers these programs reached, and do not speak to the welfare effects of forgiveness. ~~Existing evidence answers neither directly. Dinerstein et al. (2025) compare borrowers whose debt was canceled between 2021 and 2024 with borrowers who had not yet received forgiveness, and estimate the change in their borrowing and earning. Such a comparison does not capture the aggregate effect, and the programs reached specific groups of borrowers.~~ Many other proposals, from

broad-based cancellation to new repayment plans, were never implemented, so their effects cannot be measured directly.[1] To answer these questions, we develop a general equilibrium model with incomplete markets that incorporates the main institutional features of federal student loan repayment, and use it to evaluate forgiveness policies that have not been implemented. ~~the consumption stimulus and welfare effects of different relief policies.~~

Our main finding is that student debt forgiveness poses a tradeoff between stimulating the consumption of forgiven borrowers and raising welfare. ~~consumption stimulus and welfare.~~ Policies that concentrate forgiveness on low-income borrowers increase welfare but reduce the consumption response of forgiven borrowers. ~~consumption stimulus.~~ This tradeoff is surprising because a large share of low-income borrowers are liquidity constrained, and both empirical evidence and incomplete-markets models suggest that concentrating fiscal transfers on constrained households raises consumption stimulus (Oh and Reis 2012; Jappelli and Pistaferri 2014). Consistent with this intuition, cash transfers to the same borrowers do not pose the tradeoff. Student debt forgiveness generates weak consumption stimulus among low-income borrowers largely because of the repayment plans meant to protect them. Income-driven repayment (IDR) plans cap student debt payments at a fraction of labor income, and delinquent borrowers are not making payments, so for both groups a lower balance often does little to lower their current payment. ~~does not lower their current payment.~~

We begin by measuring how many constrained households carry student debt. Following Kaplan and Violante (2014), we classify households as hand-to-mouth (HtM), our measure of constrained households, using the Survey of Consumer Finances (SCF). We consider two measures: liquid HtM households carry no liquid wealth between pay periods, and net-worth HtM households carry no net worth between pay periods.[2] Kaplan and Violante (2014) abstract from student debt; we include it in illiquid wealth, by the

1. Appendix A outlines recent student debt relief proposals in the U.S.

2. A household is liquid HtM if its net liquid wealth is positive but less than half of its pay-period earnings, or if its net liquid wealth is negative and within half of its pay-period earnings of its credit limit. A household is net-worth HtM if the same criterion is applied to its net worth rather than its net liquid wealth.

same logic they apply to mortgage debt. This choice is conservative. Counting student debt as liquid debt, like a credit card balance, would classify many more borrowers as liquid HtM. Under our classification, the share of liquid HtM households that hold student debt more than doubled from 2001 to 2016, from 12% to 25%. Student debt is even more prevalent among net-worth HtM households, 49% of whom held student debt in 2016.

These stylized facts and the debate about the consumption effects ~~aggregate effects~~ of student debt relief policies lead us to develop a structural general equilibrium model with incomplete markets and uninsurable earnings risk. Households enter the workforce as college graduates, college dropouts, or high school graduates. They make consumption and savings decisions over the life cycle, saving in a single liquid asset. College graduates and dropouts may carry long-term student loan debt and repay the debt on one of two payment plans. Borrowers on the standard payment plan pay interest and a fraction of the balance each period. On the IDR plan, they pay the lesser of that amount and a fixed fraction of income. When borrowers make an income-based payment, the amount does not depend on the balance owed. Borrowers on the standard plan have the option to be delinquent, but unlike unsecured consumer debt (Livshits et al. 2007; Chatterjee et al. 2007), student loans are not discharged in bankruptcy. The balance accrues interest and rolls over (Ionescu 2009).

We discipline the model by calibrating it to match key moments on education, student debt, and repayment status. We then test it against two features of the data it was not asked to match. The first is the share of HtM households holding student debt, which the model reproduces.[3] The second is the marginal propensity to consume (MPC) out of a change in debt payments. Between 2021 and 2024, the federal government canceled the full remaining balances of roughly 7% of borrowers through a series of administrative programs. Dinerstein et al. (2025) use credit-bureau data on those borrowers to estimate an MPC of 0.57 out of the resulting change in debt payments. Running the same experiment in the model produces an MPC of 0.60. The model matches this estimate because

3. Because the model has a single asset, we compare it to the share of net-worth HtM households with student debt in the SCF.

of the share of HtM borrowers, whose MPC of 0.79 is much higher than the 0.51 of non-HtM borrowers. HtM borrowers cannot borrow against lower future obligations, so a lower balance raises their consumption mainly when it lowers their current payment.

Our baseline policy forgives up to $10,000 of student debt for every borrower earning less than $125,000.[4] Debt forgiveness is unanticipated, and the government finances it by reducing government spending. Aggregate consumption increases by 0.15% in the year of forgiveness, and the total consumption stimulus, measured as the present discounted value of the cumulative increase in consumption, is 2.3% of steady-state consumption~~, a consumption multiplier of 0.80~~. We compare this policy with cash transfers to the same borrowers who are eligible for forgiveness. Cash transfers are more commonly studied in the literature (Johnson et al. 2006; Parker et al. 2013; Kaplan and Violante 2014). We set the size of the transfer so that it requires the same reduction in government spending, in present value, as debt forgiveness. The targeted cash transfers generate a stronger initial increase in aggregate consumption, 0.45%, because the transfer arrives at once, while forgiveness lowers payments gradually over many years. Their total consumption stimulus is slightly smaller, 2.1%.

Our main quantitative exercise compares the baseline policy with four alternatives that progressively lower the income cap from $125,000 to $24,400, reducing the share of eligible borrowers from 95% to 45%. Each policy forgives the same aggregate balance of student debt, so policies with lower income caps forgive more debt per borrower for a smaller set of lower-earning borrowers. As forgiveness becomes more concentrated on low-income borrowers, the average consumption-equivalent welfare gain increases from 0.28% to 0.37% of lifetime consumption, and the total consumption stimulus falls from 2.3% to 1.8% of steady-state consumption.[5] Concentrating forgiveness therefore raises welfare at the cost of the consumption response of forgiven borrowers. Cash transfers do not pose this tradeoff: concentrating them on the same low-income borrowers raises both welfare and consumption stimulus. ~~When we instead concentrate cash transfers~~

4. This policy follows the Biden administration's original plan. The plan also had a higher threshold for eligibility for married couples, and provided up to $20,000 for Pell Grant recipients. We abstract from these additional features.

5. Welfare effects are averaged across all households, not just those who received forgiveness.

~~on the same low-income borrowers, welfare rises and consumption stimulus increases slightly.~~

Two institutional features of student debt repayment, IDR plans and delinquency, sever the link between a reduction in the debt balance and a change in payments. For borrowers on IDR plans, debt forgiveness only lowers their current payment if it lowers the standard payment below the income-based payment. This matters most at the bottom of the earnings distribution. In the first earnings quartile (Q1), 73% of forgiven borrowers are HtM, compared with 11% in the fourth quartile (Q4). To quantify how much these features limit the transmission of forgiveness to payments, we calculate the payment gap, the difference between the change in payments if every borrower were on the standard repayment plan and the actual change in payments. For Q1 borrowers under the baseline policy, forgiveness would lower payments by 8.7% of the debt forgiven if every borrower were on the standard plan, but actual payments fall by only 3.2%. IDR plans and delinquency therefore block 63% of the potential reduction in payments. Across all forgiven borrowers, IDR plans account for 70% of the payment gap and delinquency for the remaining 30%.

The decomposition shows that IDR plans account for most of the payment gap, but it does not show whether these plans are responsible for the tradeoff itself. To answer this, we compare the same five forgiveness policies in two counterfactual economies, one in which IDR plans are unavailable and one in which borrowers cannot be delinquent. When we remove IDR plans, the tradeoff disappears. Many more low-income borrowers then repay on the standard plan, where the payment is proportional to the balance, so forgiveness lowers their payments directly. Because these borrowers are the most likely to be HtM, concentrating forgiveness on them raises welfare from 0.54% to 0.82% of lifetime consumption and consumption stimulus from 3.0% to 3.4% of steady-state consumption. When we shut down delinquency, the tradeoff survives. Only borrowers on the standard plan can be delinquent, and most low-income borrowers are on IDR plans, whose payments depend on income rather than on the balance. IDR plans make

student debt manageable for low-income borrowers, but they also weaken the stimulus from forgiving their debt.

Finally, we study how the source of financing changes the effects of debt forgiveness. To the best of our knowledge, U.S. debt relief proposals have not specified how they would be financed. We decompose the aggregate consumption response into a direct effect, the change in consumption of forgiven borrowers, and indirect effects that come from the equilibrium response of prices and government policy. The direct effect does not depend on how forgiveness is financed, and our main result holds under every financing rule we consider: concentrating forgiveness on low-income borrowers lowers the consumption response of forgiven borrowers and raises welfare. Welfare gains are smaller when households pay for forgiveness, but they remain positive. The aggregate consumption response, in contrast, depends on how forgiveness is financed. When forgiveness is financed by reducing government spending, the aggregate response is close to the direct effect. When households pay through higher income taxes, the indirect effects offset nearly all of the direct effect. How the source of financing shapes the aggregate effects of fiscal transfers in economies where Ricardian equivalence does not hold remains an open question (Kaplan et al. 2018; Auclert et al. 2024; Blanchard 2019; Angeletos et al. 2024), and we do not attempt to resolve it. We focus instead on the effects of student debt forgiveness on the borrowers who receive it, which do not depend on the answer.

The rest of the paper is organized as follows. Section 2 places the paper in the literature. Section 3 estimates the share of HtM households with student debt in the SCF. Section 4 describes the model economy. Section 5 maps the model to the data. Section 6 compares the model to the MPC out of a change in debt payments estimated in the literature. Section 7 quantifies the consumption stimulus and welfare effects of debt forgiveness policies, and Section 8 concludes.

2 Related Literature

This paper is closely related to empirical work on student debt relief.[6] Dinerstein et al. (2025) study the administrative discharges of 2021–2024 and find that forgiven borrowers had higher earnings on average than other borrowers, and that forgiveness raised mortgage, auto, and credit card borrowing and reduced earnings. They also estimate an MPC of 0.57 out of the resulting change in debt payments, which we use to validate our model. Catherine and Yannelis (2023) value forgiveness by the discounted payment savings it produces and find it regressive. High earners hold larger balances, and for low earners the balance overstates those savings, because under current law much of it would be forgiven under IDR anyway. Two studies separate the balance from the payment. Di Maggio et al. (2026) study a discharge of private student debt for defaulted borrowers who had already stopped paying, which raised net wealth without changing payments, and find improvements in their credit and labor market outcomes. Dinerstein et al. (2024) study the reverse, a 2020 payment pause that suspended required payments without canceling any debt, and find that it raised borrowers' consumption. Together, these studies suggest that the effects of relief depend on whether it changes payments or only balances. We build this distinction into a general equilibrium model, which lets us compare alternative relief designs and measure their aggregate effects.

A related literature studies the two institutions that govern what a borrower actually pays. The first is IDR plans. Studies examine what raises enrollment in these plans (Mueller and Yannelis 2022), how they trade off insurance against moral hazard (de Silva 2025), and how payments under them track income fluctuations (Gervais et al. 2023). The second is delinquency. Because student loans are almost never discharged in bankruptcy (Ionescu 2009), a delinquent borrower stops paying without extinguishing the balance. The two institutions interact: Herbst (2023) finds that enrolling in IDR cuts delinquency by 22 percentage points. Two papers treat repayment rules as policy instruments in their own right. Boutros et al. (2024) show that contracts deferring payments to later in life deliver welfare gains similar to those of debt forgiveness without its fiscal cost. Catherine

6. For a broad survey, see Yannelis and Tracey (2022).

et al. (2026) find in a life-cycle model that an optimally designed IDR plan achieves the welfare gains of existing IDR plans at much lower cost to taxpayers. These papers study how repayment rules affect borrowers' repayment behavior and welfare. We ask how these rules shape the consumption response to debt forgiveness, which depends on how much forgiveness lowers borrowers' current payments.

This paper contributes to the literature on quantitative models with student debt. These models study the effects of student debt on college enrollment and default (Ionescu 2009), credit constraints and human capital accumulation (Lochner and Monge-Naranjo 2011), education policy and intergenerational transfers (Abbott et al. 2019), and job choice and wages (Luo and Mongey 2026). Alon et al. (2026) evaluate student debt forgiveness, focusing on its effects on occupational choice and early-career human capital investment. Kim and Kim (2026) model college attendance, borrowing, and delinquency, and show that the option to become delinquent substantially amplifies the rise in student debt. Our model is built to study the consumption response to debt relief. Without targeting them, it reproduces both the share of constrained households holding student debt and a micro estimate of the MPC out of changes in debt payments. We use it to measure how repayment institutions shape the aggregate consumption and welfare effects of debt relief.

This paper also speaks to the literature on how household debt affects consumption and the aggregate economy, including mortgage debt (Mian et al. 2013; Di Maggio et al. 2017; Cloyne et al. 2020; Garriga and Hedlund 2020; Kaplan et al. 2020) and unsecured credit (Gross and Souleles 2002; Herkenhoff 2019; Aydin 2022; Herkenhoff et al. 2024; Irwin 2025). Athreya et al. (2025) find that household indebtedness and a history of financial distress shape the consumption response to macroeconomic shocks. Most closely related, Ganong and Noel (2020) find that for mortgage borrowers receiving loan modifications, principal reductions that leave monthly payments unchanged have no effect on default or consumption, while maturity extensions that lower monthly payments have large effects. We show that the same distinction between balances and payments matters for student debt relief. Because IDR plans and delinquency limit how much forgiveness

lowers payments for many constrained borrowers, concentrating forgiveness on them raises welfare but lowers consumption stimulus.

Finally, this paper relates to the literature on the consumption response to fiscal transfers. Households spend a large share of tax rebates and stimulus payments (Johnson et al. 2006; Parker et al. 2013), and the response is largest among households with few liquid assets (Fagereng et al. 2021). Kaplan and Violante (2014) account for these responses with wealthy hand-to-mouth households, who hold substantial illiquid wealth but spend much of any additional cash. A common implication is that targeting transfers toward constrained households raises the consumption stimulus they generate. Using survey data, Jappelli and Pistaferri (2014) find that households with low cash-on-hand have much higher MPCs, and that transfers targeted to the bottom of the cash-on-hand distribution raise aggregate consumption substantially. Oh and Reis (2012) obtain the same result in an incomplete-markets model, in which transfers from low-MPC to high-MPC households boost aggregate demand. We show that this implication holds for cash transfers in our model but does not carry over to debt relief: concentrating forgiveness on low-income borrowers raises welfare but lowers consumption stimulus.

3 Hand to mouth households with student debt

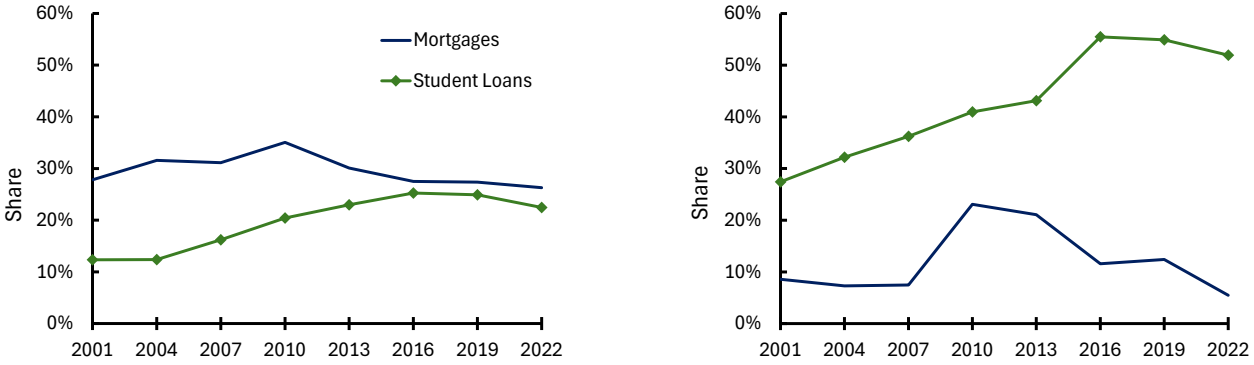
The consumption response to fiscal stimulus depends crucially on whether households are liquidity constrained (Kaplan and Violante 2014). Debt forgiveness will stimulate consumption to the extent that it reaches constrained, high-MPC households. In this section, we document the share of hand-to-mouth (HtM) households with student debt using the Survey of Consumer Finances (SCF). We show that this share has grown over time and is now comparable to the share of HtM households with mortgage debt, motivating our study of the stimulus effects of student debt relief.

Following Kaplan and Violante (2014), we compute two alternative HtM classifications. The first, liquid HtM, identifies households whose net liquid wealth is small relative to their pay-period earnings or who are at their revolving-credit limit. This is

the standard definition in the literature, since a household with illiquid wealth but no liquid savings can still have a high MPC. The second, net-worth HtM, applies the same threshold using net worth instead of liquid wealth. This is a more direct comparison to our single-asset model and is the relevant measure for student loan borrowers who cannot pay off their outstanding balances even by liquidating current assets. For both, we estimate the share of HtM households and the share of HtM households with student debt. Because we classify student loan debt as part of illiquid wealth, it does not directly affect the estimation of liquid HtM. We use SCF waves from 2001 through 2022. Full asset definitions and sample summary statistics are in Appendix C.

Figure 1(a) depicts the share of liquid HtM households with outstanding balances of student debt and mortgage debt in the SCF from 2001 through 2022. Two facts stand out. First, the share of liquidity-constrained households carrying student debt doubled from 12% in 2001 to 25% in 2016. Second, as of the 2016 SCF, the share of HtM households with student debt is nearly as large as the share with mortgage debt balances (27% in 2016).

Figure 1: Share of hand-to-mouth households with types of outstanding debt



(a) Liquid Hand-to-Mouth

(b) Net worth Hand-to-Mouth

Notes.– Share of hand-to-mouth households with student debt balances or outstanding mortgage debt. Panel (a) uses a hand-to-mouth definition based on the household's net liquid assets. Panel (b) uses a definition based on the household's net worth. Data from the Survey of Consumer Finances 2001-2022.

Student loan debt is even more prevalent among net-worth HtM households (Figure 1(b)). Over 51% of these households have student debt balances over the past three surveys, up from 27% at the turn of the century. Mortgage debt is far less common,

as between 5% and 12% of net-worth HtM households hold it in the last three SCF waves. There is a relatively small share of mortgage holders who are net-worth HtM because most have equity in a piece of real estate. This highlights an important difference between mortgage holders and student borrowers. Many mortgage holders have the option to borrow against the equity in their home, whereas most student borrowers have non-positive net worth.

Taken together, the large and growing share of constrained households with student debt makes the aggregate consumption response to student debt relief a first-order policy question. Whether this constrained population actually translates HtM status into a high MPC out of debt forgiveness, however, depends on institutional features of student loan repayment, and this question motivates our structural model.

4 Model

This section builds a general equilibrium model with three types of agents. Overlapping generations of households choose consumption, savings, and whether to default on student debt. Idiosyncratic risk and incomplete asset markets generate an endogenous distribution of households over individual state variables (Huggett 1993; Aiyagari 1994). A representative firm produces a single good using capital and two skill-differentiated labor inputs. The government collects income taxes and student loan payments and distributes transfers to households.

4.1 Households

The economy is populated by a unit mass of households. Let $j \in \{1, \ldots, J\}$ index age. Households work from age 1 through the retirement age J^R and retire thereafter. Each period, the oldest cohort exits and a new cohort of equal measure enters, so the age distribution is stationary with cohort mass $1/J$. Survival within the lifespan is certain, and all households die at the end of age J. Preferences are CRRA with risk aversion σ and discount factor β.

Households begin the life-cycle after post-secondary education. Let $e \in \{\ell, m, h\}$ denote education attainment, where ℓ denotes high school graduates, m denotes college dropouts, and h denotes college graduates. The population shares of the three education groups are π_ℓ, π_m, and $1 - \pi_\ell - \pi_m$, respectively. All households begin their lives with no savings, $a = 0$. Graduates and dropouts draw initial student debt from education-specific distributions $\mathcal{B}_0^e$. Finally, student loan borrowers ($b_0 > 0$) initially enter on the standard fixed-payment plan.

During the working years, households inelastically supply labor to the firm, where $\bar{n}$ is hours worked each period. High-skilled workers earn wage per efficiency unit of labor w^h while low-skilled workers (dropouts and high-school graduates) earn w^ℓ. Individual labor productivity is given by equation (1), where γ_j is an age–experience premium, z_p is a persistent productivity shock that follows a Markov chain with probability matrix π, and z_t is a transitory shock. After retirement, households receive social security benefits $w^e \theta^R(z)$ from the government.

$$\log y_j = \log \gamma_j + z_p + z_t \tag{1}$$

Student loan repayment begins at age 1. Households repay over N_p periods. Let p denote the number of payments made on the loan. Two repayment plans are available, a standard fixed-payment plan and an income-driven repayment (IDR) plan, indexed by $s \in \{s^f, s^{IDR}\}$. Under the fixed-payment plan, each period the borrower repays a fraction $\lambda(p)$ of the outstanding principal (equation (2)), and any unpaid balance accrues interest at the student-loan rate $r_b = r + \iota$, a fixed spread ι over the risk-free rate.[7]

$$\lambda(p) = \frac{1}{N_p + 1 - p}, \quad p = 1, \ldots, N_p \tag{2}$$

Each period, a borrower on the fixed-payment plan chooses whether to make the scheduled payment or be delinquent. Equation (3) describes this decision, where d is

7. In the US, the majority of student loans are issued by the federal government. The government sets the interest rate as a spread over the average return on 10-year Treasury notes.

the delinquency decision rule and $V_j^{e,0}$, $V_j^{e,1}$ are the values of repayment and delinquency, respectively. Borrowers incur a utility cost χ when delinquent, capturing the non-pecuniary costs of missed payments.[8]

$$V_j^e(z,a,b,p,s^f) = \max\left\{V_j^{e,0}(z,a,b,p,s^f),\ V_j^{e,1}(z,a,b,p,s^f) - \chi\right\} \tag{3}$$

$$d_j(z,a,b,p,s^f) = \begin{cases} 1 & if \quad V_j^{e,1}(z,a,b,p,s^f) - \chi \geq V_j^{e,0}(z,a,b,p,s^f) \\ 0 & otherwise \end{cases}$$

Equation (4) summarizes the required debt payment across different payment plans and delinquency choices. Under IDR, the required payment is the minimum of the fixed-plan payment and a fraction ζ^{IDR} of after-tax labor income. Delinquency is not permitted under IDR. Borrowers on the IDR plan make payments until age J^{IDR}, at which point any remaining balance is forgiven.[9,10]

$$\Lambda_j^{e,d}(z,b,p,s) = \begin{cases} \lambda(p)b & if \quad s = s^f \quad and \quad d = 0 \\ 0 & if \quad s = s^f \quad and \quad d = 1 \\ \min\{\lambda(p)b,\ \zeta^{IDR}(1-\tau)\, w^e\, \bar{n}\, y_j\} & if \quad s = s^{IDR} \end{cases} \tag{4}$$

Households optimally choose their consumption c and liquid savings a' in each model period (equation (5)).[11] Liquid savings pays interest r, and $\underline{a}$ is the borrowing limit in the liquid asset. Let τ be an income tax and T be a lump-sum transfer from the government. In the steady state $T = 0$. Transfers are used only in the cash transfer experiments in Section 7. Households making a debt payment increment p by one.

8. In the United States, student loans are considered in default after 270 days of missed payments. Unlike other consumer debt, student loans are not dischargeable under Chapters 7 or 13 of the bankruptcy code.

9. In the US, income-contingent repayment plans typically last between 20 and 25 years, after which the remaining balance is forgiven.

10. If a borrower entered retirement with student debt, the fixed payment would apply with no delinquency option. This does not occur in equilibrium. Distressed borrowers repay under IDR before reaching retirement age.

11. For households on the IDR plan, there is no delinquency decision.

$$V_j^{e,d}(z,a,b,p,s) = \max_{c,a'} \; u(c) + \beta E V_{j+1}^{e}(z',a',b',p',s') \tag{5}$$

$$\text{s.t.} \quad c + a' = (1+r)a + (1-\tau)\mathcal{E}_j - \Lambda_j^{e,d}(z,b,p,s) + T,$$

$$b' = (1+r_b)(b - \Lambda_j^{e,d}(z,b,p,s)),$$

$$a' \geq \underline{a}, \quad c \geq 0,$$

$$\mathcal{E}_j = \begin{cases} \bar{n}\, w^e\, y_j & if \quad j \leq J^R \\ w^e\, \theta^R(z) & otherwise \end{cases}$$

$$s' = \begin{cases} s^{IDR} & if \quad s = s^{IDR} \quad or \quad b > (1-\tau)\,\mathcal{E}_j\,\bar{b}^{IDR} \\ s^f & otherwise \end{cases}$$

The s' transition rule in equation (5) captures IDR enrollment. A borrower on the fixed plan is automatically switched to IDR from the next period when their current debt-to-income ratio $b/((1-\tau)\mathcal{E}_j)$ exceeds the threshold $\bar{b}^{IDR}$. Once enrolled, the transition is permanent.

4.2 The Firm

A representative firm produces output using capital K and two types of labor. High-skilled labor L^h is supplied by college graduates, and low-skilled labor L^ℓ is supplied by high-school graduates and dropouts. Aggregate labor L is a CES aggregator of L^h and L^ℓ, with elasticity of substitution ν and weight θ on high-skilled labor. The firm operates a constant returns to scale (CRS) technology in (K, L) with capital share α and depreciation rate δ, choosing inputs to maximize profits Π as in (6).

$$\Pi = \max_{K,L^h,L^\ell} \; K^\alpha L^{1-\alpha} - w^h L^h - w^\ell L^\ell - (r+\delta)K \tag{6}$$

$$where \quad L = \left[\theta (L^h)^{\frac{\nu-1}{\nu}} + (1-\theta)(L^\ell)^{\frac{\nu-1}{\nu}}\right]^{\frac{\nu}{\nu-1}}$$

4.3 The Government

The government collects income taxes $\tau \mathcal{E}_j$ and student debt payments $\Lambda_j^{e,d}(z,b,p,s)$, and pays social security payments $w^e \theta^R(z)$ and lump-sum transfers T, which are zero in the steady state. It can also borrow by issuing one-period bonds B_G, which are held by households and pay the interest rate r. In the steady state the government holds no debt, so $B_G = 0$. Government consumption G is determined residually by the budget constraint (7).

$$G = \int_{\mathcal{S}} \Big[\underbrace{\tau\, \mathcal{E}_j}_{\textit{income tax}} + \underbrace{\Lambda_j^{e,d}(z,b,p,s)}_{\textit{debt payments}} - \underbrace{\mathbf{1}[j > J^R]\, w^e\, \theta^R(z)}_{\textit{retirement benefits}} \Big] \mu(d\Psi) + B_G' - (1+r) B_G - T \quad (7)$$

where μ denotes the household distribution over the state space, formally defined in the following section.

4.4 Equilibrium Definition

Let $\Psi = (j, e, z, a, b, p, s)$ denote a household's state, where the state space is

$$\mathcal{S} = \{1, \ldots, J\} \times \{\ell, m, h\} \times \mathcal{Z} \times \mathcal{A} \times \mathcal{B} \times \{1, \ldots, N_p\} \times \{s^f, s^{IDR}\}.$$

Let μ denote the household distribution as a probability measure on the Borel σ-algebra of $\mathcal{S}$. The debt state b is set to zero for non-borrowers ($e = \ell$).

A *stationary recursive equilibrium* is a collection of value functions $\{V^e, V^{e,1}, V^{e,0}\}_{e \in \{\ell,m,h\}}$, decision rules $\{c(\Psi), a'(\Psi), d(\Psi)\}$, prices (w^ℓ, w^h, r), and a stationary distribution μ such that

1. **Optimality.** Given prices, the decision rules $c(\Psi)$ and $a'(\Psi)$ solve the household's recursive problem in equation (5). For borrowers on the fixed plan ($e \in \{m,h\}$, $s = s^f$), the delinquency rule $d(\Psi) \in \{0,1\}$ solves equation (3). For all other households $d(\Psi) = 0$.

2. **Endogenous state transitions.** The repayment-plan transition obeys

$$s'(\Psi) = \begin{cases} s^{IDR} & \text{if } b/\left[(1-\tau)\,\mathcal{E}_j\right] > \bar{b}^{IDR} \text{ and } s = s^f, \\ s & \text{otherwise,} \end{cases}$$

so that households auto-enroll in IDR when the debt-to-income ratio exceeds $\bar{b}^{IDR}$ and IDR enrollment is absorbing. Debt evolves as

$$b'(\Psi) = (1 + r_b)\left[b - \Lambda_j^{e,d(\Psi)}(z, b, p, s)\right],$$

where $\Lambda_j^{e,d}$ is defined in equation (4), and $b'(\Psi) = 0$ for non-borrowers ($e = \ell$).

3. **Firm optimality.** Prices satisfy the firm's first-order conditions for problem (6)

$$w^h = (1-\alpha) K^{\alpha} L^{-\alpha} L^{1/\nu} \theta \left(L^h\right)^{-1/\nu} \tag{8}$$

$$w^\ell = (1-\alpha) K^{\alpha} L^{-\alpha} L^{1/\nu} (1-\theta) \left(L^\ell\right)^{-1/\nu} \tag{9}$$

$$r = \alpha K^{\alpha-1} L^{1-\alpha} - \delta \tag{10}$$

4. **Government budget.** The government budget constraint (7) holds.

5. **Law of motion.** The distribution evolves according to $\mu' = \mathcal{T}(\mu)$, where the operator $\mathcal{T}$ is defined by

$$\mathcal{T}(\mu)(A) = \int_{\mathcal{S}} \mathbf{1}\left[\Psi'(\Psi, z') \in A\right] \Pi^e(z, dz')\, \mu(d\Psi) \tag{11}$$

for all measurable sets $A \subseteq \mathcal{S}$. Here $\Psi'(\Psi, z') = (j+1, e, z', a'(\Psi), b'(\Psi), p'(\Psi), s'(\Psi))$ is the next-period state implied by the decision rules and state transitions, and $\Pi^e(z, \cdot)$ is the Markov chain for the idiosyncratic productivity shock (degenerate for $j > J^R$). New cohorts enter at $j = 1$ with education drawn from population shares $(\pi_\ell, \pi_m, 1 - \pi_\ell - \pi_m)$, initial assets $a_1 = 0$, initial debt $b_1 \sim \mathcal{B}_0^e$ for borrowers ($b_1 = 0$ for $e = \ell$), and initial plan $s_1 = s^f$. At the stationary distribution, $\mu = \mathcal{T}(\mu)$.

6. **Market clearing.**

Assets.

$$\int_{\mathcal{S}} a(\Psi)\,\mu(d\Psi) = K + B_G \tag{12}$$

Households save in capital and government bonds. Because there is no aggregate risk, no arbitrage equates the returns on the two assets, so households are indifferent over the composition of their savings.

Skilled labor (college graduates).

$$L^h = \int_{\mathcal{S}} \mathbf{1}[j \leq J^R]\,\bar{n}\,y_j\,\mathbf{1}[e = h]\,\mu(d\Psi) \tag{13}$$

Unskilled labor (dropouts and high-school graduates).

$$L^\ell = \int_{\mathcal{S}} \mathbf{1}[j \leq J^R]\,\bar{n}\,y_j\,\mathbf{1}[e \in \{m, \ell\}]\,\mu(d\Psi) \tag{14}$$

Goods.

$$C + I + G = F(K, L) \tag{15}$$

5 Model to Data

We calibrate the model in three stages. First, we set a subset of parameters exogenously, fixing them at values common in the literature or estimated directly from the data. Second, we internally calibrate the remaining parameters to match the moments summarized in Table 1. Finally, we test the model against untargeted moments. Section 3 reports the share of hand-to-mouth households in the data and how closely the model reproduces it.

5.1 Externally Set Parameters

Demographics and Preferences. A model period is one year. Households are born into the model at age 22 ($j = 1$), the age at which post-secondary education is complete, work for 46 periods through age 67 ($J^R = 46$), spend the following 21 periods in retirement, and die deterministically at age 88 ($J = 67$). We set the coefficient of relative risk aversion $\sigma = 2.0$.

Wage Process. We estimate the wage process using the Panel Study of Income Dynamics (PSID) from 1968 through 2023, using individuals aged 25–65 following Kim (2022) and Heathcote et al. (2010).[12] An OLS regression of log hourly wages on time dummies, an interaction of time dummies with a college-education dummy, and a second-order polynomial in labor-market experience captures the deterministic component, namely the college wage premium and the age–earnings profile $\log \gamma_j$ in equation (1). The average college premium over the sample period is 1.8, and the experience premium yields the standard hump-shaped life-cycle earnings profile. Note that the CES weight θ is calibrated internally so that the model reproduces the premium. From the regression residuals, we estimate the idiosyncratic wage shocks which are the sum of a persistent component z_p following an AR(1) process and a transitory component z_t. The persistence and time-varying variances are estimated jointly by minimum-distance methods and each variance averaged over the sample periods.

The estimated persistence of z_p is 0.976, the variance for the persistent shock is 0.030, and that for the transitory shock is 0.087. We also set hours worked so that households spend one-third of their time working.

Government. The labor-income tax rate τ is 0.26, following Domeij and Heathcote (2004). The social security replacement rate ζ^R is 0.40, following Hosseini (2015). Thus, social security benefit amounts to 40% of average pre-retirement earnings, as defined in equation (16).

12. Details of the estimation procedure are in the appendix.

$$\theta^R(z) = \zeta^R \bar{n} \exp(z) \frac{\sum_{k=1}^{J^R} \gamma_k}{J^R} \tag{16}$$

Income-Driven Payment Plan The payment share ζ^{IDR} is 0.10, matching the roughly 10% of discretionary-income cap in current federal IDR plans, and the years of payment before forgiveness J^{IDR} is 25 years.

Production. The capital share α is 0.36 and the annual depreciation rate δ is 0.07. The elasticity of substitution between high- and low-skilled labor ν is 1.67, following Krusell et al. (2000).

5.2 Internally Calibrated Parameters

Table 1: Internally Calibrated Parameters and Targeted Moments

Parameter	Value	Target moment	Data	Model
β	0.95	Capital-to-output ratio	3.0	3.4
θ	0.576	College wage premium	1.8	1.8
ι	1.9%	Student loan interest rate wedge	2.1%	2.1%
N_p	15	Avg. student loan duration (yrs)	15	15
χ_d	0.36	Avg. delinquency rate	6.89%	6.89%
$\bar{b}^{IDR}$	1.56	Share of borrowers on IDR	42.1%	41.2%
π_ℓ	0.326	Share of college grads	39.2%	39.2%
π_m	0.282	Share of dropouts	28.2%	28.2%
π_{bh}	0.90	Share of grads with student debt	36.8%	36.4%
π_{bm}	0.90	Share of dropouts with student debt	29.8%	30.4%
μ_b^h	0.23	Avg. debt-to-income for college grads	0.87	0.89
μ_b^m	0.03	Avg. debt-to-income for dropouts	0.55	0.53
σ_b^m	0.058	Debt-to-income ratio, Q1	1.26	1.54
σ_b^h	0.058	Debt-to-income ratio, Q2	0.81	0.77

Notes. Data moments from the 2019 SCF and federal student-loan statistics.

The remaining model parameters are calibrated inside the model as shown in Table 1. We calibrate the discount factor β to target the capital-to-output ratio in the economy. The equilibrium risk-free interest rate is 3.7%.

We calibrate four parameters that govern student loan repayment and default to match features of the federal student loan program. We set the spread on the student

loan interest rate ι to 2.1%, consistent with the spread on federal student loans for undergraduate borrowers over 10-year treasury yields in the U.S. We also set the number of payments on the standard plan N_p to match the average duration of student loans at origination, 15 years. We calibrate the utility cost of delinquency χ_d to target the average delinquency rate of 6.89%. Finally, we calibrate the debt-to-income threshold for enrolling in income-driven repayment $\bar{b}^{IDR}$ to match the share of borrowers on an IDR plan, 42.1% and the model generates an IDR share of 41.2%.

We set π_ℓ and π_m so that the share of college graduates is 39% and the share of dropouts is 28%. These values correspond to the share of households with at least a 4-year degree and the share of households with some higher education. A fraction π_{bh} of college graduates and a fraction π_{bm} of dropouts draw their initial student debt from a normal distribution, $b_0 \sim \mathcal{N}(\mu_b^e, (\sigma_b^e)^2)$ for $e \in \{h, m\}$. All other households begin the life-cycle with no student debt. We set both π_{bh} and π_{bm} to 0.9 to match the share of graduates and dropouts in the data with outstanding balances of student loan debt. The means (μ_b^h, μ_b^m) are calibrated to match average student debt-to-income ratios of 0.88 for college graduates and 0.55 for dropouts. The standard deviation of the graduate debt distribution (σ_b^h) targets an average debt-to-income ratio of 0.81 for workers in the 2nd earnings quartile, and the standard deviation of the dropout debt distribution (σ_b^m) targets the debt-to-income ratio of 1.26 for workers in the 1st quartile.

5.3 Model Validation: Untargeted Moments in the Data

This section compares the calibrated model economy to untargeted moments in the data. We first examine moments over the earnings distribution of all working-age households, many of which are not directly targeted in the calibration. We then turn to the share of HtM households with student debt estimated in Section 3.

The model reproduces how education, student debt, and repayment behavior vary across the earnings distribution. Key moments in the model and the SCF are summarized in Table 2. Three patterns stand out. First, the model reproduces the steep earnings gradient across quartiles. Relative average earnings rise from 0.22 in the bottom quartile

to 2.34 in the top, close to their data counterparts of 0.27 and 2.25. Second, the share of households holding a college degree rises with income, while the share holding student debt stays relatively flat across the distribution, both consistent with the SCF. Third, and most important for our mechanism, default and debt burdens are concentrated among low-income households. The default rate falls from roughly 18% in the bottom quartile to under 1% in the top, and the average debt-to-income ratio declines from above one to well below one half. The model captures this gradient even though initial debt draws depend only on education, not on individual income realizations. Given that the disutility cost of default is also common across borrowers, the observed default and debt-burden gradients arise endogenously. Low-income borrowers face the tightest constraints and the strongest incentive to fall behind.

Table 2: The Distribution of Households Over Income

		Q1	Q2	Q3	Q4
Relative Avg. Earnings	Data	0.27	0.56	0.95	2.25
	Model	0.22	0.51	0.92	2.34
Share college grads	Data	0.16	0.31	0.43	0.68
	Model	0.25	0.25	0.54	0.54
Share with student debt	Data	0.21	0.31	0.30	0.25
	Model	0.27	0.21	0.24	0.20
Avg. debt-income	Data	1.26	0.81	0.53	0.38
	Model	1.54	0.77	0.38	0.15
Default rate (%)	Data	21.36	7.57	1.01	0.48
	Model	17.91	6.61	0.45	0.00
Share of borrowers on IDR plan	Data	0.43	0.50	0.41	0.31
	Model	0.66	0.43	0.34	0.16

Notes.– Income quartiles are constructed from the labor earnings of all working-age households (both borrowers and non-borrowers). Relative Avg. Earnings are quartile-average earnings divided by the working-age population average. Debt-income is the ratio of the student debt balance to annual labor income. Default rate is the share of borrowers with delinquent student debt. Data from the 2019 Survey of Consumer Finances (SCF).

Despite being untargeted in the calibration, the model closely matches the share of HtM households in the data (Table 3). We compare to the net-worth definition of HtM

because the model has a single liquid asset for savings. The model generates 12.9% of households as HtM, against 14.1% in the 2022 SCF. The model also generates 6.5% of households who are both HtM and student loan borrowers, versus 7.3% in the data. Matching this share matters because it shows the model generates the right proportion of constrained, high-MPC households who are repaying student debt. These households are presumably the ones who can potentially show the largest consumption response to a change in the stream of student debt payments.

Table 3: Share of HtM households in the model and data

	Data	Model
Share of HtM	14.1%	12.9%
Share of HtM with student debt	7.3%	6.5%

Notes.– Share of hand-to-mouth households based on net worth. The second row is the share of all households who are both hand-to-mouth and student loan debtors. Data from the Survey of Consumer Finances 2019.

6 The MPC out of Student Debt Payments

We study the student debt forgiveness policies of 2021 to 2024 through the lens of our model. These programs discharged the full remaining balance of over 3 million borrowers, roughly 7% of those outstanding in 2020.[13] We compare the model's consumption response with the estimate of Dinerstein et al. (2025), who study the same

13. The programs were a set of narrower administrative reforms, pursued after the Supreme Court struck down the Biden administration's broad forgiveness plan in *Biden v. Nebraska* (600 U.S. 477, 2023). Appendix A details the programs and eligibility.

programs. Using a difference-in-differences design,[14] they estimate an MPC of 0.57 out of the reduction in student loan payments due that forgiveness produces.[15]

We design the experiment in the model to match these policies. Beginning from the steady-state equilibrium, we simulate a one-time cancellation of student debt in which the probability of receiving forgiveness depends on the age of the borrower. As in Dinerstein et al. (2025), the average borrower receiving forgiveness is 12 years out of school, with a standard deviation of 5 years.[16] In total, 6.5% of borrowers in the steady state receive full forgiveness, close to the 7% share discharged by the actual programs.[17]

Table 4 reports the average MPC among borrowers who receive forgiveness and whose payment due falls. We measure the MPC as the change in consumption in the year of forgiveness divided by the change in the payment due that year. The model predicts an MPC of 0.60, close to the estimate of 0.57 in Dinerstein et al. (2025), even though this moment is not targeted in the calibration. Because the consumption effects of forgiveness in our model depend on how much forgiveness changes debt payments, this match lends support to the mechanism behind the results that follow. The rest of this section first shows how the option to be delinquent shapes the response, and then how the response differs with whether borrowers are hand-to-mouth and with their repayment plan.

14. Their main sample begins in January 2021. They estimate the average treatment effect on the treated over the 5 months before and 6 months after a forgiveness event, and find similar results with an instrumental variable estimation. Because one in four forgiven borrowers expected cancellation before it occurred, they scale the change in debt payments by 0.75, since forward-looking borrowers would already have adjusted consumption in anticipation.

15. They also report an MPC of 0.27 when the change in consumption is divided by the forgiven balance annuitized over the borrower's remaining life expectancy. We compare the model to the payment-based estimate because it divides by the same object as our MPC, the change in the payment due. Their MPC divides the increase in monthly payments due on mortgage, auto, and credit card debt by the reduction in monthly student loan payments due, with both effects pooled over the six months after forgiveness. The numerator covers only credit-financed spending, so it excludes purchases paid with cash or debit cards. It is also a payment flow on new debt, which proxies for the consumption of durables rather than measuring nondurable consumption. Our model period is a year and has one consumption good, so our MPC is the same ratio in annual flows. The two ratios are comparable if both flows persist over the year. Dinerstein et al. (2025) report that the effects on credit outcomes appear quite stable out to 12 months after forgiveness (their Appendix Table C.4).

16. In the TransUnion data, the average borrower receiving forgiveness graduated in 2009.

17. The government funds the cancellation by adjusting government consumption. Prices are held at their steady-state values. We solve for the partial equilibrium response because the difference-in-differences estimation nets out general equilibrium effects.

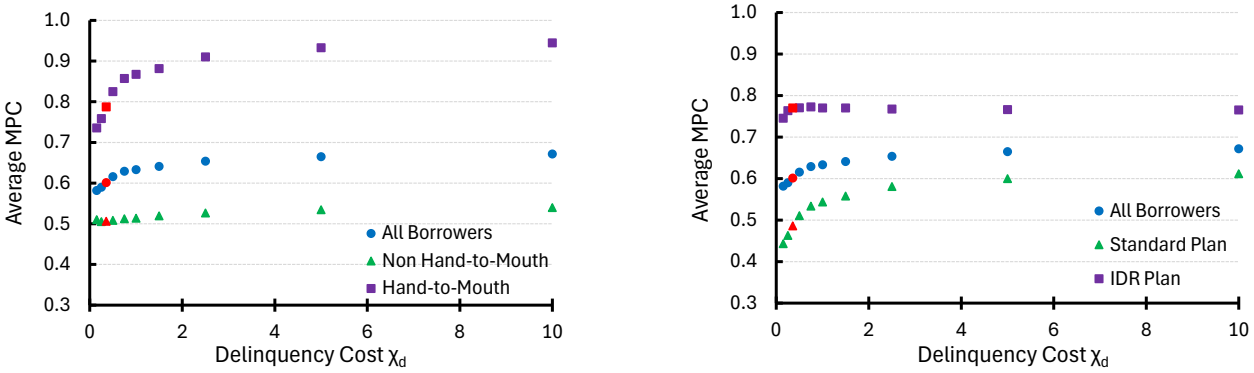
Table 4: MPC out of Change in Debt Payment

	All Borrowers	HtM	Non-HtM	Standard Plan	IDR Plan
Data	0.57	NA	NA	NA	NA
Baseline	0.60	0.79	0.51	0.49	0.77
No Delinquency	0.79	1.02	0.69	0.80	0.77

Notes.– MPC out of a change in debt payments, averaged over borrowers who receive forgiveness and whose payment due falls. Data is an estimate from TransUnion data in Dinerstein et al. (2025). Baseline is the calibrated model with $\chi_d = 0.36$. No Delinquency is a model in which borrowers have no delinquency option ($\chi_d = 10^6$). Hand-to-mouth (HtM) status is based on net worth, as in Section 3.

We begin with the option to be delinquent. Figure 2 shows how the average MPC varies with the delinquency cost χ_d. Raising χ_d from its baseline value of 0.36 to 10 cuts the delinquency rate from 6.89 percent to 0.18 percent and raises the average MPC from 0.60 to 0.67. The MPC then rises further to 0.79 in the economy in which delinquency is unavailable (Table 4). These increases reflect two channels. The first is actual delinquency. A delinquent borrower does not pay what is due, so cancelling the payment due frees little cash and consumption barely changes. As fewer borrowers are delinquent, the average MPC rises.

Figure 2: MPC by Delinquency Cost



(a) MPC by borrower type

(b) MPC by repayment plan

Notes.– MPC out of a change in debt payments for different values of the delinquency cost χ_d, which the horizontal axis labels "Default Cost." Panel (a) splits borrowers by hand-to-mouth status and panel (b) by repayment plan. The baseline ($\chi_d = 0.36$) is depicted in red. The economy without a delinquency option is reported in Table 4.

The second channel is the option itself. At $\chi_d = 10$ almost no one is delinquent, yet the average MPC of 0.67 remains well below 0.79.[18] The gap does not come from composition, since the share of HtM households (12.8 versus 12.5 percent) and the share of borrowers on IDR (39.3 versus 39.2 percent) barely change. Delinquency helps borrowers smooth consumption, since they can skip a payment after bad shocks. Without this option, borrowers must pay in every state, so a lower payment matters more for their consumption. Delinquency is therefore an important margin shaping the consumption response. In an economy without the option, the MPC of 0.79 lies well above the estimate of 0.57.

The response also differs sharply between HtM and non-HtM borrowers. Figure 2(a) shows the average MPC by borrower type. HtM borrowers have an MPC of 0.79. They hold little net worth and cannot borrow against future relief, so they spend most of the cash that a lower payment frees up. Non-HtM borrowers have a lower but still sizable MPC of 0.51.[19] Figure 2(a) also shows that the MPC of both groups rises as delinquency becomes costlier, and that the MPC of HtM borrowers rises more.

The response also depends on the borrower's repayment plan. Figure 2(b) shows that the average MPC is 0.49 for borrowers on the standard plan and 0.77 for borrowers on IDR (Table 4). The gap arises because only borrowers on the standard plan can be delinquent. The MPC of IDR borrowers stays between 0.75 and 0.77 at every value of χ_d in the figure. This is what we should expect, since IDR borrowers cannot be delinquent and the cost of delinquency does not enter their decisions.[20]

18. Table 4 also reports an MPC of 1.02 for HtM borrowers in the economy without delinquency. An MPC above one is possible because the denominator is only this year's payment, while forgiveness also cancels future payments.

19. This may seem large compared to the MPC of non-HtM households out of a one-time tax rebate. For example, the MPC of non-HtM households in the model of Kaplan and Violante (2014) is 7 percent in the quarter of receipt of an unexpected $500 rebate. The two MPCs measure different responses. A rebate is paid once, while forgiveness cancels all of a borrower's remaining payments. Our MPC divides the change in consumption by this year's payment alone, so the denominator understates how much the borrower's resources change.

20. An alternative explanation is mechanical. By construction, the IDR payment never exceeds the fixed-plan payment on the same balance, so dividing by a smaller payment could inflate the IDR ratio even if the consumption response were the same. The economy without delinquency rules this out. IDR payments are still smaller there, yet the standard-plan MPC of 0.80 is no lower than the IDR MPC of 0.77.

These MPCs describe how consumption responds when a borrower's payment falls. However, they are silent on how much forgiveness lowers payments. Forgiveness delivers little payment relief to two groups of borrowers. A lower balance reduces an IDR borrower's payment only if the fixed payment on the new balance falls below the income-based payment. As discussed above, a lower payment due also frees little cash for delinquent borrowers. Both groups are concentrated at the bottom of the earnings distribution. Among forgiven borrowers in the lowest earnings quartile, 66 percent are on IDR and 18 percent are delinquent (Table 6). The next section shows how this limited payment relief shapes the aggregate consumption and welfare effects of forgiveness.

7 The Aggregate Effects of Debt Forgiveness Policies

Section 6 showed that borrowers spend a large share of a reduction in their debt payments. This section studies how student debt forgiveness changes payments across borrowers, and what this implies for its consumption stimulus and welfare effects. We find that concentrating forgiveness on low-income borrowers raises welfare but lowers consumption stimulus, a tradeoff that does not arise with targeted cash transfers.

7.1 Consumption Stimulus and Welfare Effects

The main quantitative exercise of this paper studies student debt forgiveness policies that differ in how concentrated they are across the earnings distribution of borrowers. Two policy parameters determine eligibility and the size of forgiveness. These are the income cap and the maximum amount forgiven. Any borrower with annual earnings at or below the income cap is eligible, and each eligible borrower's debt balance is reduced by the lesser of that balance and the maximum amount.

The baseline policy we consider forgives up to $10,000 for any borrower with annual earnings at or below $125,000, following the Biden administration's original plan (Appendix A). Debt forgiveness is an unanticipated, one-time intervention. The government finances it by adjusting government consumption in equation (7), so households

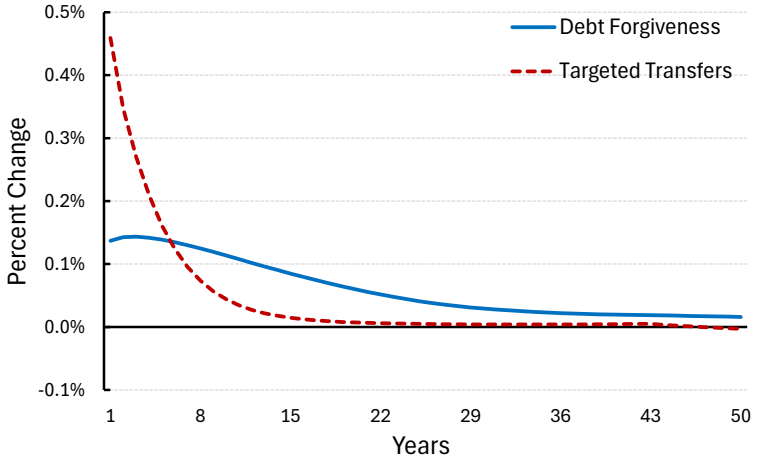
do not bear the cost of the policy through their own budgets. Section 7.4 studies how other sources of financing change the results. Consumption increases by 0.15% in the first year following forgiveness (Figure 3). The total consumption stimulus, measured as the present discounted value of consumption deviations from steady state, is 2.3% of steady-state consumption.[21] This corresponds to a consumption multiplier of 0.80. Each dollar of debt forgiven raises the present value of consumption by 80 cents.[22]

To benchmark debt forgiveness against cash transfers, a policy more commonly studied in the literature (Johnson et al. 2006; Parker et al. 2013; Kaplan and Violante 2014), we consider transfers targeted to the same borrowers. The baseline cash transfer has the same income cap of $125,000. Each eligible borrower receives an unanticipated, one-time transfer equal to 78% of the debt that would have been forgiven, financed in the same way as forgiveness. The 78% factor is the present discounted value of debt forgiveness per dollar of face value. It is below one because, in our model, many borrowers on IDR plans make income-based payments that fall below the interest on their balances, so part of the forgiven debt would never have been repaid. Scaling the transfer by this factor ensures that the two policies have the same present-value cost to the government. Figure 3 compares the aggregate consumption responses. Cash transfers raise consumption more on impact, by 0.45% in the first year, because the transfer arrives at once, while forgiveness lowers payments gradually over many years. However, the total consumption stimulus of cash transfers is slightly smaller, at 2.1% of steady-state consumption.

To trace how the consumption stimulus and welfare gains depend on how concentrated debt forgiveness is across the earnings distribution, we study five policies with income caps ranging from $125,000 to $24,400. We choose the caps so that the share of eligible borrowers falls from 95% to 45%. For each cap, we set the maximum amount to hold the total debt forgiven fixed across policies: 40% of outstanding student debt balances in the economy. Table 5 reports the resulting parameters and the share of bor-

21. Throughout, we compute present discounted values using the equilibrium risk-free interest rate.
22. The consumption multiplier is the present discounted value of the increase in consumption divided by the aggregate balance of debt forgiven.

Figure 3: Consumption Response



Notes.— Percent deviation of aggregate consumption from steady state. Periods are years. Debt forgiveness cancels up to $10,000 of debt for borrowers with annual earnings at or below $125,000. Targeted transfers pay the same borrowers a one-time transfer equal to 78% of their forgiven amount, so that both policies have the same present-value cost to the government. Both policies are unanticipated and financed by adjusting government consumption.

rowers eligible under each policy. As the income cap falls, the policies forgive a larger amount per borrower to a smaller group of lower-earning borrowers.

Table 5: Parameters of Debt Forgiveness Policies

Income Cap	Max Amount	Total Forgiven	Eligible Borrowers
$125,000	$10,000	0.40	0.95
$73,500	$11,400	0.40	0.85
$44,800	$15,000	0.40	0.70
$30,700	$22,200	0.40	0.55
$24,400	$32,200	0.40	0.45

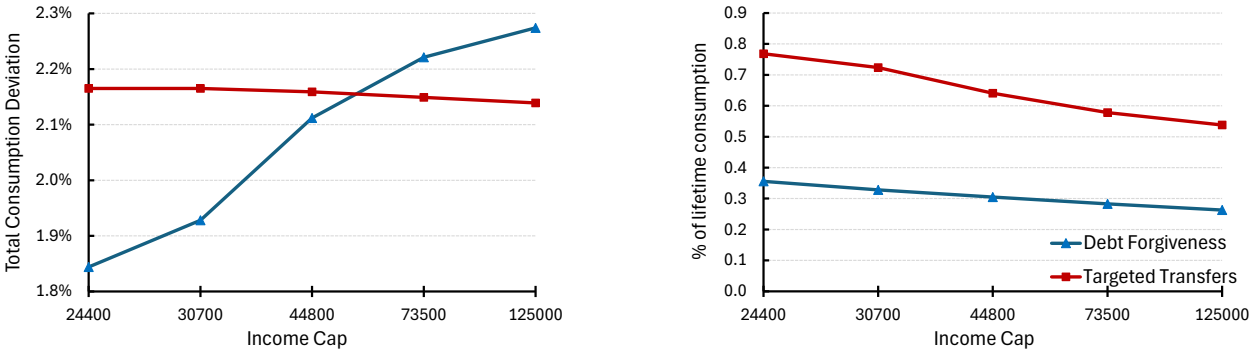
Notes.— The first row is the baseline policy. Income cap is the maximum annual earnings at which a borrower is eligible for forgiveness. Max amount is the maximum forgiveness per borrower, chosen to hold the total forgiven fixed across policies. Total forgiven is the debt forgiven as a share of aggregate student debt balances. Eligible borrowers is the share of all borrowers eligible for forgiveness.

Figure 4 plots the consumption stimulus and welfare effects of each forgiveness policy, along with those of the cash transfers discussed below. We measure welfare as the consumption-equivalent gain averaged across all households, which is the share of lifetime consumption they would give up to receive the policy. We find that student debt

forgiveness poses a tradeoff between welfare and consumption stimulus. As forgiveness becomes more concentrated on low-income borrowers, the stimulus effect falls and the welfare gain rises at each step. Moving from the broad-based policy capped at $125,000 to the concentrated policy capped at $24,400, consumption stimulus declines from 2.3% to 1.8% of steady-state consumption, while the average welfare gain to households rises from 0.28% to 0.37% of lifetime consumption.

We run the same experiment with targeted cash transfers. For each of the five income caps, eligible borrowers receive a one-time transfer equal to the lesser of their student debt balance and a maximum amount. We set the maximum amount so that the aggregate transfer is the same across policies, at 2.0% of steady-state output. Cash transfers do not pose the same tradeoff between welfare and consumption stimulus. As they become more concentrated on low-income borrowers, both the consumption stimulus and the welfare gain rise. Moving from the broad-based policy capped at $125,000 to the concentrated policy capped at $24,400, the consumption stimulus increases slightly, while the average welfare gain rises from 0.55% to 0.78% of lifetime consumption.

Figure 4: Consumption and Welfare Response to Debt Forgiveness and Targeted Transfers



(a) Aggregate Consumption

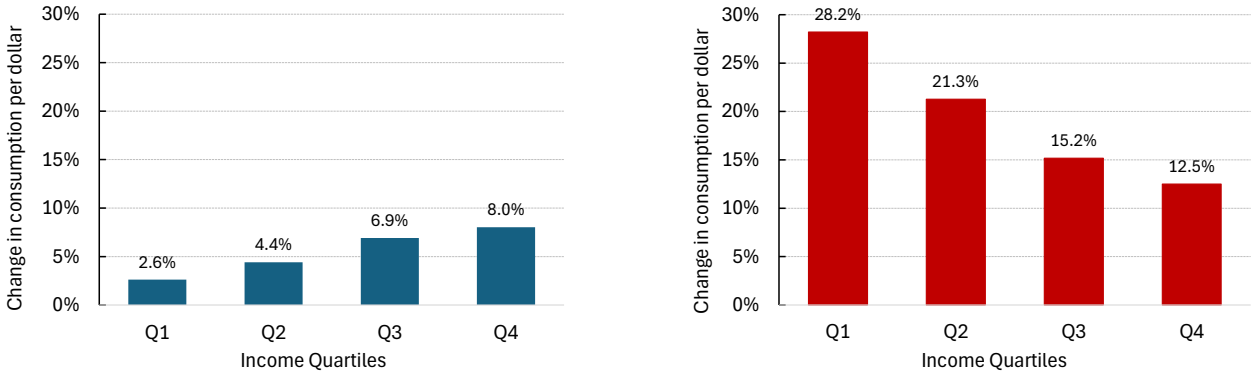
(b) Average Welfare

Notes.— Each point is a policy with the income cap shown on the horizontal axis. The parameters of the debt forgiveness policies are reported in Table 5. Targeted transfers use the same income caps, and the aggregate transfer is 2.0% of steady-state output under every cap. Panel (a) reports consumption stimulus, the present discounted value of aggregate consumption deviations from steady state, as a percent of steady-state consumption. Panel (b) reports welfare, the consumption-equivalent gain averaged across all households, as a percent of lifetime consumption.

The difference in the pattern of consumption stimulus between the two policies reflects a weaker response to debt forgiveness among low-income borrowers. Figure 5 reports the change in consumption in the first year of the policy per dollar forgiven or transferred, by earnings quartile, for the broad-based policy. With debt forgiveness, the response is smallest for low-income (Q1) borrowers, at 2.6% of the amount forgiven, and rises monotonically to 8.0% for high-income (Q4) borrowers. The ordering reverses with targeted cash transfers. Q1 has the largest response, at 28.2% of the amount transferred, and the response declines steadily up the earnings distribution to 12.5% in Q4. The weak response to forgiveness among low-income borrowers reflects that it lowers their payments little, because most of them are on IDR plans or delinquent (Section 7.2).

Welfare, by contrast, falls with earnings under both policies. Figure 6 reports the welfare gain by earnings quartile for the same policies, averaged over forgiven borrowers rather than all households. Forgiveness is worth 1.9% of lifetime consumption to Q1 borrowers and 0.5% to Q4 borrowers, while the transfer is worth 4.9% to Q1 and 0.6% to Q4.

Figure 5: Consumption Response Across Earnings Distribution



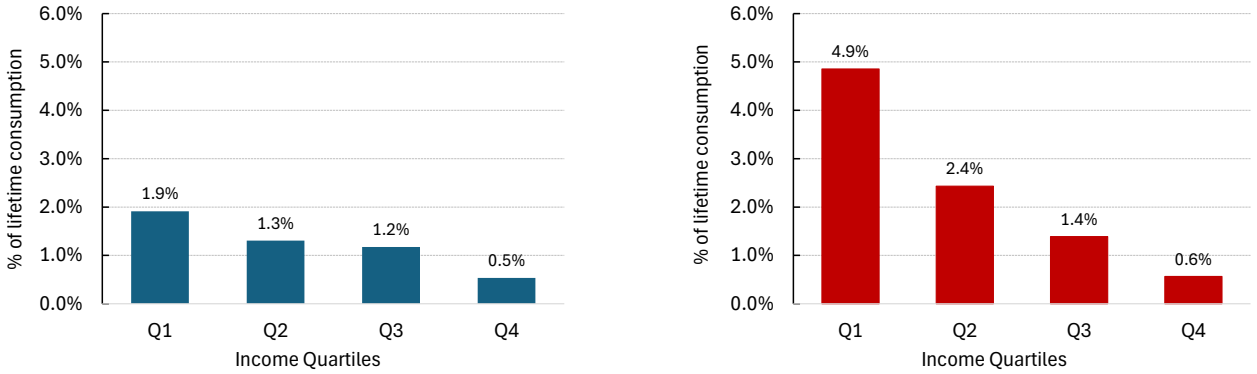
(a) Debt Forgiveness

(b) Targeted Transfers

Notes.— Change in first-year consumption per dollar forgiven or transferred, by quartile of the earnings distribution, in percent. Panel (a) shows the baseline debt forgiveness policy, which forgives up to $10,000 for borrowers with annual earnings at or below $125,000. Panel (b) shows the baseline cash transfer policy, which pays the same borrowers a one-time transfer equal to 78% of their forgiven amount. Responses are lower in panel (a) in part because forgiveness lowers payments over many years, while a transfer arrives at once.

The tradeoff between consumption stimulus and welfare has a direct implication for the design of student debt forgiveness. The choice of the income cap depends on the

Figure 6: Welfare Response Across Earnings Distribution



(a) Debt Forgiveness (b) Targeted Transfers

Notes.— Consumption-equivalent welfare gain, in percent of lifetime consumption, averaged within each quartile of the steady-state earnings distribution of forgiven borrowers. Panel (a) shows the baseline debt forgiveness policy, which forgives up to $10,000 for borrowers with annual earnings at or below $125,000. Panel (b) shows the baseline cash transfer policy, which pays the same borrowers a one-time transfer equal to 78% of their forgiven amount. Both policies are financed by adjusting government consumption.

policy's goal. Within the set of policies we study, broad-based forgiveness generates the most consumption stimulus, while concentrating forgiveness on low-income borrowers generates larger welfare gains. Targeted cash transfers do not face this tradeoff. Concentrating them on low-income borrowers raises welfare without lowering consumption stimulus. The difference arises because the borrowers most responsive to cash are the least responsive to debt forgiveness.

7.2 The Transmission Mechanism

Debt forgiveness generates a smaller consumption response among low-income borrowers because it reduces their outstanding balance but often leaves their current payment unchanged. For borrowers on IDR plans, the payment is the smaller of the scheduled payment and a cap set at 10% of income. As long as this income-based cap binds, forgiveness leaves the payment unchanged, and it lowers the payment only once the scheduled payment falls below the cap. Delinquent borrowers are not making payments, so forgiveness cannot lower their current payment. In both cases, forgiveness raises the borrower's net worth but does little to relax the current budget constraint. A

cash transfer, by contrast, relaxes the current budget constraint regardless of repayment status.

Table 6 shows why these frictions matter most for low-income borrowers. Among forgiven borrowers in Q1, 66% are enrolled in IDR plans and 18% are delinquent, leaving only 16% on the standard plan. The pattern reverses at the top. In Q4, 83% are on the standard plan and the remaining 17% are on IDR. Low-income borrowers are also far more likely to be hand-to-mouth, 73% in Q1 against 11% in Q4, and they have the highest MPC out of a change in debt payments, 0.65 against 0.58 in Q4. Low-income borrowers are therefore the most likely to spend payment relief. Yet they are exactly the borrowers whose payments forgiveness barely changes, because most of them are on IDR or delinquent.

Table 6: Eligibility and Payment Status by Income Quartile

	Q1	Q2	Q3	Q4
Eligible for forgiveness (%)	100.0	100.0	100.0	73.4
Among forgiven borrowers				
IDR enrolled (%)	65.6	42.5	33.6	17.0
Delinquent (%)	17.9	6.6	0.4	0.0
Fixed payment (%)	16.4	50.8	65.9	83.0
Hand-to-mouth share (%)	72.6	47.5	27.9	10.8
MPC debt payment	0.65	0.56	0.59	0.58

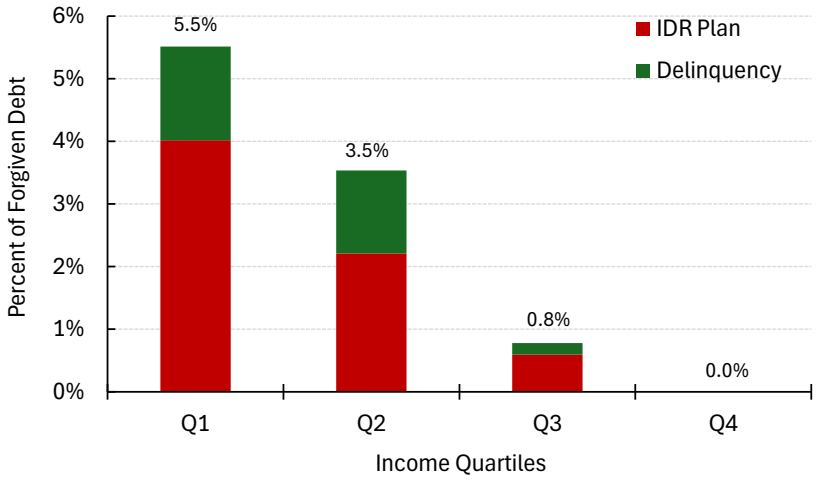
Notes.— Statistics by quartile of the steady-state earnings distribution of borrowers under the baseline debt forgiveness policy, which forgives up to $10,000 for borrowers with annual earnings at or below $125,000. Eligible for forgiveness is the share of all borrowers in each quartile with earnings at or below the income cap. All other rows are computed among forgiven borrowers. IDR enrolled, delinquent, and standard plan are mutually exclusive repayment statuses. Hand-to-mouth households are defined as in Section 3. MPC debt payment is the average marginal propensity to consume out of a change in debt payments.

To quantify how much IDR plans and delinquency limit the transmission of forgiveness to current payments, we compute the potential change in payments. This is the first-year change in payments if every forgiven borrower were repaying on the standard

plan. We define the payment gap as the difference between the potential and actual changes in payments. It measures how much IDR plans and delinquency reduce the payment relief that forgiveness delivers. Figure 7 plots the payment gap per dollar of forgiven debt across the earnings distribution of borrowers.

Forgiveness does much less to lower the payments of low-income borrowers. The payment gap for Q1 borrowers is 5.5% of forgiven debt.[23] Their consumption rises by only 2.6% of forgiven debt (Figure 5(a)). With an MPC of 0.65 out of a change in debt payments, IDR plans and delinquency hold back a large part of their potential consumption response.[24] The payment gap falls in each quartile as earnings rise. In Q4 it is zero, because every forgiven borrower is either on the standard plan or on an IDR plan whose payment already equals the standard payment.

Figure 7: Payment Gap by Earnings Quartile



Notes.— Payment gap by quartile of the earnings distribution of forgiven borrowers under the baseline debt forgiveness policy. The potential change in payments is the first-year change in payments if every forgiven borrower were repaying on the standard plan. The actual change in payments is the first-year change in payments given borrowers' repayment status. The payment gap is the potential minus the actual change, as a percent of forgiven debt. Each bar splits the gap into the part from borrowers on IDR plans and the part from delinquent borrowers. Labels show the total.

23. The potential change in payments for Q1 borrowers is 8.7% of forgiven debt and the actual change in payments is 3.2%.

24. A simple back-of-the-envelope calculation suggests that the consumption response of Q1 borrowers would increase from 2.6% of forgiven debt to 6.2% given the MPC and the change in potential payments. However, this would be an upper bound for the change in consumption because IDR borrowers have a higher MPC on average than borrowers on standard plans (see Section 6).

IDR plans account for most of the payment gap, but delinquency also accounts for a sizable share. In Q1, where the gap is largest, IDR plans account for 73% of the gap and delinquent borrowers for the remaining 27%, although only 18% of forgiven borrowers in Q1 are delinquent. Across all forgiven borrowers, IDR plans account for 70% of the gap and delinquency for 30%.

These frictions explain why concentrating forgiveness on low-income borrowers lowers consumption stimulus even as it raises welfare. As the income cap falls, forgiveness goes to low-income borrowers, most of whom are on IDR plans or delinquent. Less of each dollar forgiven then reaches current payments, and consumption stimulus declines. Welfare still rises because forgiveness raises these borrowers' net worth, which is most valuable to low-income borrowers. Because most of them are hand-to-mouth and cannot borrow against lower future debt obligations, their consumption responds to current payments rather than to net worth. A cash transfer relaxes the current budget constraint directly, so concentrating it on low-income borrowers raises welfare without lowering consumption stimulus.

7.3 Counterfactual Economies without Delinquency and IDR

Section 7.2 showed that IDR plans and delinquency limit how much debt forgiveness lowers current payments, particularly for low-income borrowers. That decomposition holds borrowers' behavior fixed. To test whether these frictions explain the tradeoff between consumption stimulus and welfare, we quantify the effects of the policies in Table 5 in two counterfactual economies. The first has no IDR plans, so borrowers repay on the standard plan or become delinquent. The second has no option to become delinquent. All other parameters are held at their baseline values, and borrowers re-optimize in each economy. Figure 8 reports the consumption stimulus and welfare effects of the five policies in each economy.

Removing IDR plans eliminates the tradeoff. Without IDR plans, many low-income borrowers become delinquent instead. In Q1, 53.2% of borrowers are delinquent, compared with 17.9% in the baseline. But the rest repay on the standard plan, where the

payment is proportional to the balance, so forgiveness lowers their payments directly. The share of Q1 borrowers whose payments fall with forgiveness therefore rises from about 16% to 47%, and the reduction in their payments rises from 3.2% to 6.1% of the debt forgiven. These borrowers are the most likely to be hand-to-mouth, and they have the highest MPC out of a change in payments. Concentrating forgiveness on them therefore raises both consumption stimulus and welfare, as it does with targeted cash transfers. Moving from the broad-based policy capped at $125,000 to the concentrated policy capped at $24,400, consumption stimulus rises from 3.0% to 3.4% of steady-state consumption, and the average welfare gain rises from 0.54% to 0.82% of lifetime consumption.

Removing delinquency, by contrast, leaves the tradeoff in place. Only borrowers on the standard plan can be delinquent, and the lowest-income borrowers are mostly on IDR plans, whose payments depend on income and not on the balance. Without delinquency, these borrowers remain on IDR, so forgiveness still does little to lower their payments. As in the baseline economy, consumption stimulus falls from 2.2% to 1.8% and the welfare gain rises from 0.33% to 0.42% as forgiveness becomes more concentrated on low-income borrowers.[25] This is consistent with IDR plans accounting for most of the payment gap in Section 7.2.

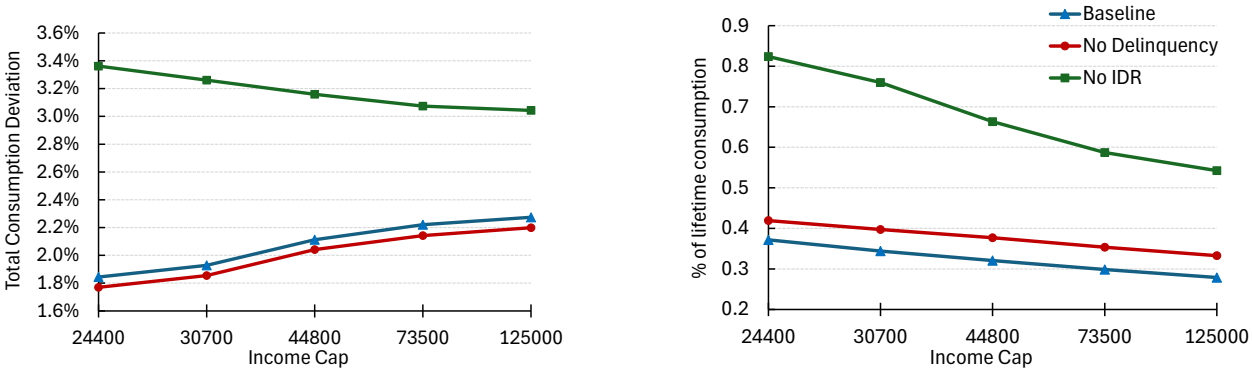
Concentrated forgiveness targets exactly the borrowers whose payments IDR plans already limit, so it adds little payment relief for them. This does not imply that IDR plans are undesirable, since they insure borrowers against periods of low income.

7.4 The Source of Financing for Debt Forgiveness

In the previous subsections, the government finances debt forgiveness by adjusting government consumption, so households do not bear the cost of the policy through their own budgets. We now study how the source of financing affects the consumption stimulus and welfare effects of debt forgiveness. To do so, we implement a financing

25. Without delinquency, the MPC out of a change in debt payments is higher (Table 4), but borrowers also repay faster, so forgiveness cancels smaller balances. The two effects largely offset in aggregate consumption.

Figure 8: Consumption and Welfare Response in Counterfactual Economies



(a) Present Value Consumption Deviation

(b) Average Welfare

Notes.— Each point is a debt forgiveness policy with the income cap shown on the horizontal axis. Policy parameters are reported in Table 5. *Baseline* is the economy described in Section 4. *No IDR* is an economy without IDR plans, in which borrowers repay on the standard plan or become delinquent. *No Delinquency* is an economy without the option to become delinquent. All other parameters are held at their baseline values, and borrowers re-optimize in each economy. All policies are financed by adjusting government consumption. Panel (a) reports consumption stimulus, the present discounted value of aggregate consumption deviations from steady state, as a percent of steady-state consumption. Panel (b) reports welfare, the consumption-equivalent gain averaged across all households, as a percent of lifetime consumption.

rule that determines how much of the policy is financed with government debt and how much with income taxes, and how quickly the government repays any accumulated debt.

Government consumption is now held at its steady-state level. When the government forgives student debt, it runs a deficit $\Delta_t^G < 0$, determined by the right-hand side of equation (7). Equations (17) and (18) describe how the government finances this deficit. A share ϕ^G is financed with debt, and the remaining share $1 - \phi^G$ is financed by adjusting the income tax rate τ_t, where $\mathcal{E}_t$ denotes aggregate taxable income in period t.[26] Households hold government debt, which pays the same return as capital, r_t. Government debt therefore crowds out capital. The government repays its debt over time by adjusting the tax rate, repaying a share ϕ^B of outstanding debt each period, so that government debt returns to its steady-state level in the long run.[27]

26. Aggregate taxable income is the wages paid to low-skilled and high-skilled labor plus social security benefits.

27. We study an economy with no government debt in steady state, so government debt converges to zero under all of the financing rules we consider.

$$B_{G,t+1} = B_{G,t} - \phi^G \Delta_t^G - \phi^B \left(B_{G,t} - \overline{B}_G \right) \tag{17}$$

$$(\tau_t - \overline{\tau})\, \mathcal{E}_t = -(1 - \phi^G)\Delta_t^G + \phi^B \left(B_{G,t} - \overline{B}_G \right) \tag{18}$$

Adding the two equations gives the government budget constraint, in which the deficit is covered by new borrowing and additional tax revenue. We consider three financing rules that differ in how aggressively the government adjusts taxes to finance debt forgiveness, and compare each with the expenditure-financed benchmark from the previous subsections. Under the first two, the government initially finances the full deficit with debt ($\phi^G = 1$) and repays it over time by raising taxes. The rules differ in how quickly the debt is repaid. Under *slow-repayment debt financing*, we set $\phi^B = 0.067$, so roughly one-fifteenth of accumulated debt is repaid each period. Under *fast-repayment debt financing*, we set $\phi^B = 0.2$, so one-fifth is repaid each period.[28] Under *tax financing*, we set $\phi^G = 0$, so forgiveness is financed entirely by adjusting taxes in the period it occurs, and no debt accumulates.

For each financing rule, we decompose the aggregate consumption response into a direct effect and indirect effects. The direct effect is the change in consumption of forgiven borrowers relative to what they would have consumed without forgiveness, given the path of prices and taxes. It therefore does not depend on how forgiveness is financed because the source of financing only affects consumption to the extent if affects prices and taxes. The indirect effects are the change in consumption of all households associated with the equilibrium response of prices and government policy following forgiveness. Both are measured as the present discounted value of consumption deviations, as a share of steady-state aggregate consumption. Table 7 reports the direct effect, the aggregate consumption response, and the welfare effects for each debt forgiveness policy in Table 5.

28. For both debt-financing rules, we set $\phi^B = 0.5$ after 50 years to ensure that the economy converges to steady state by the end of the transition. The results are nearly identical when we instead solve the model over a longer transition without changing ϕ^B.

Table 7: Consumption and Welfare Effects under Alternative Financing Rules

	Income cap				
	$24,400	$30,700	$44,800	$73,500	$125,000
Panel A. Direct consumption effect					
All financing rules	1.87	1.96	2.16	2.28	2.34
Panel B. Aggregate consumption response (direct + indirect)					
Expenditure financing	1.84	1.93	2.11	2.22	2.27
Slow-repayment debt financing	0.058	0.059	0.060	0.059	0.059
Fast-repayment debt financing	0.058	0.060	0.061	0.060	0.060
Tax financing	0.059	0.060	0.061	0.061	0.060
Panel C. Welfare, forgiven borrowers					
Expenditure financing	1.66	1.54	1.45	1.36	1.28
Slow-repayment debt financing	1.56	1.43	1.33	1.23	1.15
Fast-repayment debt financing	1.54	1.42	1.31	1.21	1.12
Tax financing	1.53	1.41	1.30	1.20	1.11
Panel D. Welfare, all households					
Expenditure financing	0.37	0.34	0.32	0.30	0.28
Slow-repayment debt financing	0.32	0.29	0.26	0.23	0.21
Fast-repayment debt financing	0.29	0.26	0.23	0.20	0.18
Tax financing	0.28	0.25	0.21	0.18	0.15

Notes.— Each column is a debt forgiveness policy with the income cap shown in the column heading. Policy parameters are reported in Table 5. Under *expenditure financing*, the government adjusts government consumption. Under *slow-repayment* and *fast-repayment debt financing*, the policy is financed with government debt ($\phi^G = 1$) that is repaid by adjusting the income tax rate, with $\phi^B = 0.067$ and $\phi^B = 0.2$, respectively. Under *tax financing*, the deficit is financed each period by adjusting the income tax rate ($\phi^G = 0$). Panels A and B report the present discounted value of consumption deviations from steady state, as a percent of steady-state aggregate consumption. The direct effect in Panel A is the change in consumption of forgiven borrowers relative to what they would have consumed without forgiveness, and does not depend on the financing rule. The aggregate response in Panel B also includes the indirect effects of the equilibrium response of prices and government policy. Panels C and D report the consumption-equivalent welfare gain as a percent of lifetime consumption, averaged across forgiven borrowers and across all households, respectively.

Table 7 shows that the source of financing has little effect on the direct consumption response, but a large effect on the aggregate response. Because the direct effect does not depend on the financing rule, the consumption of forgiven borrowers increases by 2.34% of steady-state consumption under the broad-based policy capped at $125,000, regardless of how the policy is financed. With expenditure financing, the aggregate response of 2.27% is close to the direct effect, because the reduction in government consumption frees resources that households consume. When households instead bear the cost

through higher taxes, the indirect effects offset nearly all of the direct effect: the aggregate response is about 0.06% of steady-state consumption under both debt-financing rules and under tax financing, for every policy we consider. The small aggregate response reflects the resource constraint of the economy. With government consumption fixed, forgiveness transfers resources between households and the government without changing the resources available to the economy. Any short-run increase in consumption comes at the expense of investment, and so of output and consumption in later periods, which leaves the present value of the aggregate consumption response close to zero.

The welfare effects of debt forgiveness are less sensitive to the source of financing than the aggregate consumption response. Welfare gains are smaller when households pay for forgiveness through taxes, but they remain positive under every financing rule. For forgiven borrowers, the welfare gain from the broad-based policy falls from 1.28% of lifetime consumption with expenditure financing to between 1.11% and 1.15% when households bear the cost. Across all households, the gain falls by more in proportion, from 0.28% to between 0.15% and 0.21%, because households without student debt pay part of the taxes without receiving forgiveness. In both cases, the gain is smallest under tax financing, where households pay soonest. Forgiveness remains welfare improving because it redistributes resources toward households who value them most: student borrowers are on average younger and hold fewer liquid assets than the average taxpayer.

The welfare ranking of policies does not depend on the source of financing. Under every financing rule, concentrating forgiveness on lower-income borrowers raises welfare, for both forgiven borrowers and all households. Moving from the broad-based policy to the concentrated policy capped at $24,400 lowers the direct consumption effect from 2.34% to 1.87% of steady-state consumption, but raises the welfare gain for all households from 0.15% to 0.28% under tax financing, and from 0.28% to 0.37% under expenditure financing.

Under tax and debt financing, the aggregate response is close to zero for every policy, so concentrating forgiveness raises welfare without lowering aggregate consumption. As for other fiscal policy (Auclert et al. 2024) and monetary policy (Kaplan et al. 2018), the

aggregate effects of forgiveness depend on how it is financed, and how financing shapes these effects is debated (Blanchard 2019; Angeletos et al. 2024). Like Kaplan and Violante (2014), who study the household-level consumption response to the 2001 tax rebates, we are primarily interested in how forgiveness changes the consumption of the households who receive it. Under every financing rule, this direct effect falls as forgiveness becomes more concentrated, reflecting how IDR plans and delinquency limit the payment relief that forgiveness gives to low-income borrowers.

8 Conclusion

This paper asks how student debt relief affects consumption and welfare, and how the design of relief shapes those effects. We study these questions in a general equilibrium model of federal student loan repayment that matches the estimated marginal propensity to consume out of payment reductions in recent forgiveness programs.

Our main finding is that student debt forgiveness poses a tradeoff between consumption stimulus and welfare. Concentrating forgiveness on low-income borrowers raises welfare but lowers consumption stimulus, even though these borrowers are the most likely to be hand-to-mouth. Cash transfers to the same borrowers do not pose this tradeoff. The difference comes from how student debt is repaid. Many low-income borrowers are on IDR plans or are delinquent, so a lower balance does not lower their current payment, and hand-to-mouth borrowers respond mainly to payments rather than to balances. Removing IDR plans eliminates the tradeoff, while removing delinquency does not. As with other fiscal transfers, the aggregate consumption response also depends on how forgiveness is financed, but concentrating forgiveness raises welfare and does not raise consumption stimulus under any financing rule we consider.

These results do not imply that forgiveness or IDR plans are undesirable. Forgiveness raises average welfare under every policy we consider, and IDR plans tie payments to income, which insures borrowers against periods of low income. But a policy aimed at raising consumption among low-income borrowers would do more by raising their

current resources directly, as cash transfers do, than by reducing balances that IDR plans already limit. Borrowers spend a large share of any reduction in their payments (Section 6), so what matters for their consumption is how much relief reaches their payments.

Several extensions remain. We are agnostic about whether borrowers are constrained because of debt overhang or would have been constrained anyway because of other forces, such as life-cycle dynamics or preference heterogeneity. Distinguishing the two is a promising direction for future work. The empirical literature also highlights a link between student debt forgiveness and other forms of household debt, such as mortgages, auto loans and credit cards. A model that integrates multiple forms of household debt could clarify the mechanisms behind this link and its implications for student debt reform, and we pursue this in a separate paper that studies the implications of allowing student loans to be discharged in bankruptcy. The model also holds labor supply fixed, whereas Dinerstein et al. (2025) find that forgiveness reduced earnings. Allowing labor supply to respond could change the aggregate effects and is another direction for future work. Finally, the aggregate state of the economy may affect how debt relief reaches consumption. During Covid-19, the U.S. and Canada both froze debt payments for student borrowers, making student debt relief a natural candidate for counter-cyclical policy in future recessions. We leave the analysis of how these effects differ across expansions and recessions for future research.

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A Student Debt Relief in the United States

Growing debt balances and high default rates have led to numerous proposals about how to deal with student loan debt. Many proposals have been made and some have been implemented. This section provides background information on student loan debt in the U.S. and details many of the proposals that have been made to provide student debt relief. Dinerstein et al. (2025) provides a more detailed summary of recent policies that have been implemented to provide debt relief.

Traditionally, it has been very difficult for borrowers to discharge student loan debt in default. Default is classified as 270 days of missed payments. It is quite common. For cohorts graduating between 1995 and 2017, between 15% and 28% of undergraduate borrowers defaulted within three years of graduation (Congressional Budget Office (2020)). Default can trigger consequences such as wage garnishment and a lower credit score. Loans in default are often restructured, and many borrowers transition to an income-driven repayment (IDR) plan. However, the underlying debt is not typically discharged or forgiven. Bankruptcy offers little relief either. Only about 0.1% of student loan debtors who file for bankruptcy each year discharge their debt (Iuliano (2020)).[29]

Borrowers with federal student loans have the option to enroll in an IDR plan. These plans are readily used. Over 40% of borrowers in the U.S. are on an IDR plan.[30] Borrowers on these plans pay a fraction of their income each month for 20 to 25 years. At the end of the plan, any remaining debt is forgiven. In principal, these plans offer a path to debt forgiveness, but in practice it was very unusual for borrowers to have any debt forgiven. Before 2020, only 50 borrowers had debt forgiven as part of an IDR plan (U.S. Department of Education (2025)).

The growing balance of student loan debt and the high default rates has resulted in an active political discussion about student loan repayment programs and debt forgiveness. Most notably, the Biden administration announced a broad-based plan for debt

29. Approximately 250,000 student loan debtors file for bankruptcy each year. Less than 300 discharge their student loans in bankruptcy.

30. Data from the 2019 SCF.

forgiveness in August of 2022. Under this proposal, the majority of federal student loan borrowers would receive between $10,000 and $20,000 in debt forgiveness.[31] The plan was challenged in court, and in June 2023 the Supreme Court ruled in *Biden v. Nebraska* that the policy exceeded the Secretary of Education's authority under the HEROES Act, blocking its implementation before any debt was canceled.[32]

Following the Supreme Court's decision, the Biden administration pursued debt relief through a series of administrative reforms to existing forgiveness programs rather than broad-based cancellation. These included a payment count adjustment for IDR plans, a temporary waiver loosening eligibility requirements for the Public Service Loan Forgiveness (PSLF) program, the introduction of the Saving on a Valuable Education (SAVE) plan, an expansion of borrower defense discharges, and regulatory changes to total and permanent disability (TPD) discharges.[33] Collectively, these adjustments resulted in debt cancellation for about 3 million borrowers, which is approximately 7% of the all individuals with student loans in 2020. For most borrowers, the remaining balance of their student loan debt was forgiven. Dinerstein et al. (2025) show that on average, forgiven borrowers were older and had higher income than unforgiven borrowers.

Federal policies regarding student loan repayment go beyond debt forgiveness programs. In response to the economic disruption caused by the Covid-19 pandemic, both the United States and Canada suspended federal student loan payments. In the U.S., the Coronavirus Aid, Relief, and Economic Security (CARES) Act suspended interest accrual and monthly payments on federally-held student loans as part of a broader bipartisan package of economic relief measures.[34] The pause was extended multiple times by both the Trump and Biden administrations before finally concluding in September 2023, making it one of the longest-running provisions of pandemic-era economic stimulus. The

31. Eligible borrowers earning less than $125,000 annually (or $250,000 for married couples) would receive up to $10,000 in debt cancellation, with Pell Grant recipients eligible for up to $20,000. The administration estimated that more than 40 million borrowers would qualify for this plan.

32. *Biden v. Nebraska*, 600 U.S. 477 (2023)

33. See Dinerstein et al. (2025) for a detailed summary on all of the changes that led to an increase in student debt forgiveness.

34. Canada similarly suspended federal student loan payments and interest under the COVID-19 Emergency Response Act.

inclusion of the student loan payment pause within broad economic stimulus legislation makes clear that policymakers viewed the relief as a tool for economic stabilization, not merely as targeted assistance to student borrowers.

B Additional details for parameterization

Table B1: Externally Set Parameters

Description	Parameter	Value
Risk aversion coefficient	σ	2.00
Hours worked	$\bar{n}$	0.33
Capital share	α	0.36
Capital depreciation rate	δ	0.07
Elasticity of sub., skilled/unskilled labor	ν	1.67
Labor income tax rate	τ	0.26
Social security replacement rate	ζ^R	0.40
Persistence of productivity	ρ^e	0.976
Variance, persistent shock innovation	σ_η^2	0.030
Variance, transitory shock	σ_t^2	0.087
Number of periods	J	67
Retirement age	J^R	46
IDR payment share of income	ζ^{IDR}	0.10
Periods until IDR forgiveness	J^{IDR}	25

C SCF Data and HtM Classification Methodology

C.1 SCF sample and asset definitions

We estimate the share of hand-to-mouth (HtM) households using the Survey of Consumer Finances (SCF), a cross-sectional survey of the assets and debts of U.S. households. For consistency with the literature, we follow the methodology in Kaplan and Violante (2014). We drop the top 5% of households by net worth. Liquid assets are defined as cash, money market, checking, savings, call accounts, mutual funds, bonds, and T-Bills net of credit card balances.[35] Illiquid assets are defined as housing, retirement accounts, life insurance policies, CDs, and saving bonds net of mortgage debt, home equity lines

35. We use the same cash imputation procedure as Kaplan and Violante (2014).

of credit, and student loan debt. This is the same definition for illiquid wealth used in the literature with one important difference. We account for student loan debt.

C.2 Summary statistics

Table C1 details summary statistics for households with and without student debt balances in the 2019 SCF. Student debt holders are younger and have higher labor income. This is not surprising because the majority of student debt holders are college graduates, and college graduates earn nearly twice as much as non-graduates in the SCF. They also have less wealth, holding $\$1,202$ less net liquid assets (median) and $\$78,520$ less net illiquid assets (median) than households without student debt. Student debt balances are large relative to wealth. The median balance of student loan debt is roughly 10 times larger than the median liquid wealth of student debt holders, and it is roughly 55% larger than median illiquid wealth.

Table C1: Summary statistics for SCF Households

	With Student Debt		Without Student Debt	
	Median	Mean	Median	Mean
Age	37	40	55	53
Labor income	\$65,000	\$85,488	\$52,000	\$73,527
Net liquid wealth	\$2,598	\$17,367	\$3,800	\$41,673
Net illiquid wealth	\$14,880	\$90,238	\$93,400	\$229,949
Student debt balance	\$23,000	\$41,119	\$0	\$0

Notes.– Age is the age of the head of household. Drop the top 5% of households by net worth. Data from the 2019 Survey of Consumer Finances.

C.3 HtM classification and wealth definitions

Our baseline definition of HtM households uses a two-point classification. A household is considered HtM if it does not carry wealth across pay periods or if it is borrowing constrained in revolving credit. A complication with using the SCF is that it reports average values of assets and debts in a month, so we cannot directly determine whether a respondent holds wealth at the end of a pay period. Following Kaplan and Violante

(2014), a household is HtM if its average balance of wealth is less than half of its pay-period earnings. Similarly, a household is credit constrained if its available credit (credit limit minus wealth) is less than half of its pay-period earnings.

We compute two HtM classifications that differ in the wealth concept applied to the threshold. The first uses net liquid wealth, which directly translates to a definition of liquidity-constrained households. This is a standard definition in the literature because a household with illiquid wealth can still have a high MPC if it holds no liquid savings and liquidating assets is costly. The second definition uses net worth. We include the net worth definition for two reasons. First, net worth is a more direct comparison to our model where households can only save in one asset. Second, this provides a definition of constrained households that is relevant for the conversation surrounding student loans, since student borrowers cannot pay off their outstanding debt balances even if they liquidated all of their current assets. For both definitions, we estimate the share of HtM households and the share of HtM households with student debt balances. Because we classify student loan debt as part of illiquid wealth, it will not directly affect the estimation of liquid HtM.